

KOLESA.KZ

kolesa



Premium Auto Market Analysis

Kazakhstan High-End Vehicle Market · Kolesa.kz

INF 323 — Final Project · Data Analysis

1,200+

Raw Listings

60M+

KZT Threshold

9

Cities Covered

5

SQL Queries

Project Overview

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Introduction

Website selection & project goals

02



Data Collection

Web scraping with Selenium & BeautifulSoup

03



Preprocessing

Cleaning, transforming & feature engineering

04



Database Design

SQLite schema with normalized tables

05



Data Analysis

5 SQL queries with business insights

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Visualization

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Conclusion

Key findings & limitations

01 Introduction

KOLESA.KZ

Website

Kolesa.kz

Kazakhstan's largest automotive marketplace

Target Segment

Cars listed above 60,000,000 KZT — the Premium & Luxury tier of Kazakhstan's auto market.



Price Anomaly Detection

Compare actual vs. mean market price to identify undervalued or overpriced luxury assets



Regional Trends

Analyze concentration of premium listings across cities: Almaty, Astana, and beyond



Depreciation Analysis

Measure how mileage and age impact vehicle value in the 60M+ KZT bracket

02 Data Collection

TOOLS USED

Selenium

Browser automation & dynamic content loading

BeautifulSoup

HTML parsing & data extraction

Requests

Site availability checks & HTTP

Pandas

Data aggregation & CSV export

SCRAPING PROCESS

- 1 Set price filter: $\geq 60,000,000$ KZT
- 2 Scroll & paginate through listing pages
- 3 Extract card data (title, price, city, year)
- 4 Visit each listing for detail fields
- 5 Parse mean_price from window.digitalData
- 6 Export: 1,200 rows $\rightarrow$ Kolesa_Raw_Data.csv

📌 Challenges: Anti-bot detection $\rightarrow$ randomized delays (1.5–3.5s) · Dynamic JS content $\rightarrow$ Selenium wait conditions · Rate limiting $\rightarrow$ exponential backoff

03 Data Preprocessing

1,200

Rows Before

~750

Rows After

3

Cols Removed

4

Cols Added



Handle Missing Values

- Dropped 3 columns >91% null
- Filled mileage NaN → 0 (brand new)
- Filled mean_price with median



Remove Duplicates

- Step A: Exact URL-based deduplication
- Step B: Content deduplication (title + year + price + city)



Fix Data Types

- Price: string → integer (strip ₹ symbol)
- Mileage: string → integer
- Engine volume: extract float



Clean Text

- Strip whitespace & hidden chars
- Normalize to Title Case for consistent GROUP BY in SQL



Feature Engineering

- brand: first word of title
- price_diff: mean_price - price
- is_bargain: price < 95% of mean



New Computed Fields

- car_age = 2026 - year
- price_per_year = price / (age + 1)
- Output → Kolesa_Final_Cleaned.csv

04 Database Design

brands	
id	INTEGER PK
name	TEXT UNIQUE

-1 -- N

cars (main)	
id	INTEGER PK
title	TEXT
year	INTEGER
price	INTEGER
mileage	INTEGER
engine_volume_num	REAL
transmission	TEXT
bodywork	TEXT
color	TEXT
mean_price	REAL
price_diff	REAL
is_bargain	TEXT
	INTEGER

-1 -- N

cities	
id	INTEGER PK
name	TEXT UNIQUE

DESIGN DECISIONS

Normalization

brands & cities extracted to separate tables — prevents redundancy and enables fast GROUP BY JOINS

Foreign Keys

brand_id and city_id link cars to dimension tables — ensures referential integrity

Derived Cols

price_diff, is_bargain, car_age computed at load time — accelerates analytical queries

SQLite Engine

Lightweight, zero-configuration — ideal for academic analysis via Python's sqlite3 module

05 Data Analysis — SQL Queries

Q1 Average Price by Brand

```
SELECT b.name, AVG(c.price) as Avg_Price
FROM cars c JOIN brands b ON c.brand_id=b.id
GROUP BY b.name HAVING COUNT(c.id)>5 ORDER BY 2
DESC
```

→ Ferrari & Rolls-Royce lead at 160M+ KZT average — 3× more than Mercedes-Benz entry models

Q2 Listings Distribution by City

```
SELECT ci.name, COUNT(c.id) as Listings
FROM cars c JOIN cities ci ON c.city_id=ci.id
GROUP BY ci.name ORDER BY 2 DESC
```

→ Almaty (51%) + Astana (24%) = 75% of all premium inventory in Kazakhstan

Q3 Top High-End Listings

```
SELECT title, price, year, mileage
FROM cars WHERE price>50M AND mileage<20000
ORDER BY price DESC LIMIT 10
```

→ Identifies 'unicorns' — near-new super-luxury cars; 2024 models dominate the top 10

Q4 Bargain Hunting by City

```
SELECT ci.name, COUNT(*), AVG(price_diff)
FROM cars c JOIN cities ci ON c.city_id=ci.id
WHERE is_bargain='Yes' GROUP BY ci.name
```

→ Certain regional markets (Karaganda, Shymkent) show higher below-average pricing

Q5 Inventory Age Distribution

```
SELECT CASE WHEN year>=2020 THEN 'New'
            WHEN year>=2010 THEN 'Modern' ELSE 'Older' END
as Era,
COUNT(*), AVG(price) FROM cars GROUP BY Era
```

→ 76% of listings are 2020+ models; rare classics (pre-2010) maintain high collector value

06 Visualization & Dashboard



Google Sheets

Interactive pivot tables and charts built directly from Kolesa_Cleaned_Version_Sheets.xlsx for quick ad-hoc analysis.

- Dynamic pivot tables
- Conditional formatting for bargains
- City breakdown charts



Looker Studio

Full business intelligence dashboard with real-time filtering by city, brand, and price range.

- City pie chart (Almaty 50.8%)
- Brand avg-price bar chart
- Year-price trend line
- Sortable data table

KEY DASHBOARD INSIGHTS

GEOGRAPHY

50.8% Almaty · 23.7% Astana → 74.5% market concentration in 2 cities

BRAND PEAK

Ferrari & Rolls-Royce avg > 160M KZT — 2–3× higher than Porsche & Mercedes-Benz

VALUE TRAP

Pre-2010 models spike above 100M KZT — rare classics defy typical depreciation curve

07 Conclusion

KEY FINDINGS



Market is geographically concentrated — Almaty & Astana account for ~75% of all 60M+ KZT listings



Ferrari, Rolls-Royce, Bentley & Lamborghini form an ultra-premium cluster averaging above 130M KZT



Majority (76%) of premium listings are 2020+ models with 0 km, indicating a healthy new-car market



Pre-2010 cars spike in price — collectors' vehicles defy standard depreciation curves



~15% of listings qualify as 'bargains' (priced $\geq 5\%$ below market mean), mainly in secondary cities

LIMITATIONS

1. Snapshot data — listing prices fluctuate daily
2. ~37% records dropped as duplicates/outliers
3. mean_price sourced from Kolesa's own algorithm, not independent valuation
4. No historical time-series — single scrape prevents trend analysis over time

This project demonstrates an end-to-end data pipeline: web scraping → preprocessing → SQL database → BI visualization — applied to Kazakhstan's premium automotive sector.